

A Comparative Study of Multivariate Machine Learning for 10-Year Cardiovascular Disease Risk Prediction: The Effect of Feature Scaling

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Abstract: Cardiovascular disease is very common causing lots of life around the globe. As much as it's fatal and common, it can be predicted very much with the proper, regular and timely care. With the proper record keeping of the regular checkup, lifestyle, dietary, and family history data, the risk of CVD can be predicted whether the person is on risk or not within 10-year time frame, enabling ones to take the strict and proper health care approached and changes in lifestyles, potentially saving millions of lives worldwide. This study compares seven different classical machine learning algorithms for predicting ten-year coronary heart disease risk on the Framingham dataset, and also evaluates and compares the effect of scaling the continuous type of data under a leakage-controlled protocol. Logistic Regression provided the best result among other 6 method with the CV AUC of 0.725 and test AUC of 0.693 and recall of 0.605 and the scaling of feature affected significantly only SVM, did much better feature unscaled. Notably, Logistic Regression also outperformed both ensemble methods (Random Forest and Gradient boosting) indicating that the dataset weak, largely linear signal does not benefit from the added model complexity.

Keywords—Cardiovascular Heart Disease, classical ML, feature scaling, classification, Farmingham, class imbalance, SMOTE, comparative study

I. INTRODUCTION

Cardiovascular disease is one the leading cause of death worldwide, especially due to its nature. It is highly preventive but have very less time to act upon when it hits the person. So studying the nature, gathering the live data and studying upon that data can have minimize the early risk enabling the prevention of it. The prediction task here is to classify the individual for the risk of Coronary heart disease (CHD) within 10-years of time using the baseline demographic and clinical features. This is a future outcome prediction binary classification problem done using different classical ML. The analysis uses a publicly redistributed teaching subset data on kaggle derived from a access controlled Framingham dataset, a long running cohort study on patients maintained by the NHLBI.

Class imbalance is the central to the task. The target class is highly imbalance with only 644 of the 4238 observation having a positive outcome. This is heavily effect the classifier accuracy, a classifier that predicts the negative class for every observation would attain the accuracy of approximately 84.8% so the model evaluation must be considered using the ROC-AUC, recall, precision, and F1-score rather than the accuracy alone

The research question here is which of the seven machine learning algorithm provides the best prediction of the ten year

CHD under the specific pipeline and how does the standardizing the continuous data affect the model performance. The study compares logistic regression, Naïve Bayes, K-nearest neighbors, decision tree, Support Vector Machine(SVM), two ensembles – random forest and gradient boosting under a leakage-safe, imbalance aware protocol-uniform SMOTE with cross validation. The main comparison was done that uses the standardized continuous inputs and the second case without the standardization while retaining everything same, allowing us to assess the effect of standardization of data for different algorithm, which finding are later discussed on the paper.

II. RELATED WORK

Evaluating supervised machine learning algorithms on public heart disease datasets has become a standard practice in cardiovascular informatics. Researchers typically benchmark a suite of classic classifiers such as K-Nearest Neighbors (KNN), Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), and Logistic Regression (LR) to find the best predictive pipeline for clinical use [1], [3]. Yet, the literature is filled with conflicting results. For instance, while some studies claim that basic Random Forests easily reach near-perfect accuracy [3], [4], others show that tree-based models and SVMs fail on imbalanced distributions [2], [5]. When these models are tuned properly, architectures like XGBoost often squeeze out better performance than optimized linear baselines [2]. A major issue, however, is that most of these comparisons focus purely on headline accuracy [8]. By chasing a single percentage, researchers ignore vital clinical metrics like sensitivity and specificity, leaving the actual real-world reliability of these models highly questionable.

Because clinical heart data is usually messy and skewed toward healthy patients, modern pipelines try to balance the scales using data augmentation and robust scaling. Applying the Synthetic Minority Oversampling Technique (SMOTE) has become the go-to method for lifting minority class recall and F1-scores [4], [8]. But a close look at the literature reveals a widespread flaw: data leakage. Many studies apply SMOTE or normalize their features globally across the entire dataset before splitting it into train and test sets [4], [7]. This implicitly bleeds future test-set information into the training phase, leading to highly inflated, unrealistic accuracy scores. To get around this, recent work has pushed for strictly leakage-controlled pipelines [6]. Under this setup, the data is partitioned first, and all subsequent operations like imputing missing values, scaling

features with robust estimators, and generating synthetic samples happen strictly inside the training folds [6]. This ensures that the validation and test sets remain completely untouched and clinically realistic.

No matter how accurate modern machine learning models claim to be, actual clinical practice remains anchored in traditional epidemiological risk scores. The gold standard is still the Framingham General Cardiovascular Risk Profile developed by D'Agostino et al. [9]. This classical approach uses sex-specific Cox proportional-hazards regression to estimate a patient's 10-year risk of a cardiovascular event based on a simple linear combination of standard risk factors such as age, blood pressure, cholesterol levels, smoking status, and diabetes. While these regression-based models offer total clinical transparency and form the backbone of primary care guidelines, they assume rigid, linear relationships and fail to capture complex, non-linear interactions between risk factors. For computer scientists, the Framingham model serves as the essential baseline; any new machine learning system must prove its worth against this established clinical benchmark to justify its extra complexity.

III. DATASET AND EXPLORATORY ANALYSIS

A. Dataset Description

For the dataset of this project, Framinham "Cardiovascular Study" dataset was used, which have 4238 records with 16 distinct columns one target column "TenYearCHD" comprising of total 15.2% of the positive class. This data was publicly available at Kaggle which is the subset of NHLBI's anonymized Framingham teaching dataset. This subset redistributed via the Kaggle is not the full Framingham Heart Study, the main set if access controlled and is provided from the official site of the NHLBI. The features and the details of the data are shown of Table I.

The target column data is strongly imbalanced showing the 3594 rows of class 0 84.80% of the whole data and 644 15.20% of the whole data on the class 1. This dataset showed no any duplicates data, but the missing values where there on 7 out of 16 columns. 582 rows around, 13.7% of the data were affected , from which glucose being the largest value missing counts (388) as shown on the Figure I and II.

TABLE I Dataset data dictionary

Features	Meaning	Types	Units
male	Sex (1 = male, 0 = female)	Binary	—
age	Age	Continuous	years
education	Education level (1 lowest ... 4 highest)	Categorical	—
currentSmoker	Current smoker (1 = yes)	Binary	—
cigsPerDay	Cigarettes per day	Continuous	count

BPMeds	On blood-pressure medication (1 = yes)	Binary	—
prevalentStroke	History of stroke (1 = yes)	Binary	—
prevalentHyp	Hypertensive at baseline (1 = yes)	Binary	—
diabetes	Has diabetes (1 = yes)	Binary	—
totChol	Total cholesterol	Continuous	mg/dL
sysBP	Systolic blood pressure	Continuous	mmHg
diaBP	Diastolic blood pressure	Continuous	mmHg

Upon looking the column data types, we can see and have categorized on four different groups of data category as listed on Table II. This was done so that later on the preprocessing and pipeline creation we can fill the missing values , impute those data, find the best strategy for the feature scaling and the class imbalance problem.

TABLE II Dataset predictor groups and types of data category

Group	Variables
Continuous or Numeric (8)	age, cigsPerDay, totChol, sysBP, diaBP, BMI, heartRate, glucose
Binary(6)	Male, currentSmoker, BPMeds, prevalentStroke, prevalentHyp, diabetes
Ordinal category(1)	Education, recorded as levels 1-4
Target	TenyearCHD, coded 0 and 1

B. Exploratory Data Analysis

Before creating the pipeline for the data to be fed to the model, Exploratory Data Analysis (EDA) was performed which showed the characteristics of the data and helped understand the data much better revealing some important insights to the data. The target column of our data for the model "TenYearCHD" which shows the imbalance on the data of 15.2% positive motivated to look at the recall , F1 and AUC evaluation metrics over the accuracy alone, which helped us to evaluate the model much better. After checking the distribution of the continuous feature it was found out that most of them are right-skewed on which glucose was the most

skewed with the value of 6.21. Getting this skewness on the data, it help to justify the median imputation is best of the continuous data, also some outliers were found which was clinically real. With the EDA it was also found that the most feature related to the positive CHD was prior stroke (44%), diabetes (37%), on BP meds (33%) hypertension 25% and smoking barely mattered, matching with the types of diagnosis a doctor will make as shown on Figure IV. The correlation between the data features were weak with the target, max was the age and sysBP with 0.225 and 0.216 showing no any dominant predictor. Strong correlation were seen between rather other features: sysBP and diaBP with 0.78 and somker-cigsPerDay with 0.77, showing no direct relation with the target as shown on the Figure V. EDA showed the data carries a real but weak, largely linear signal.

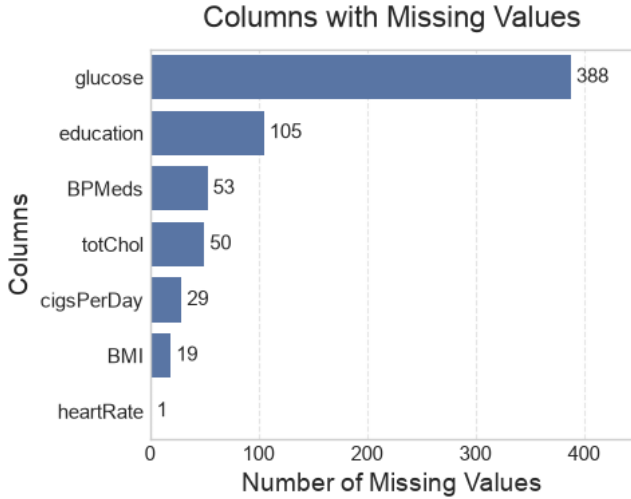


Figure I Missing values counts for each feature

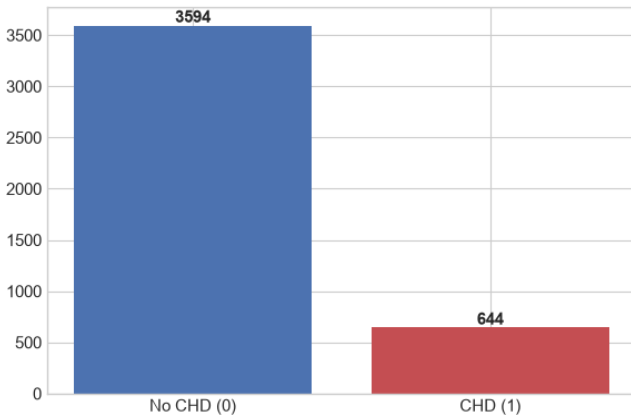


Figure II Target class binary values

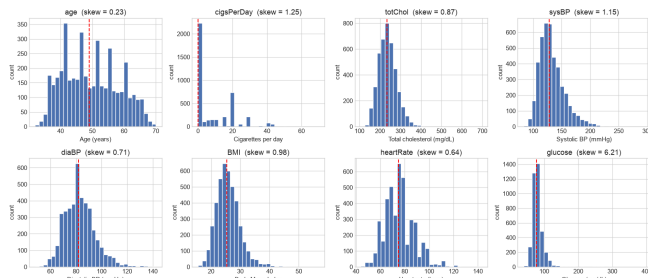


Figure III Feature distribution of the continuous type data

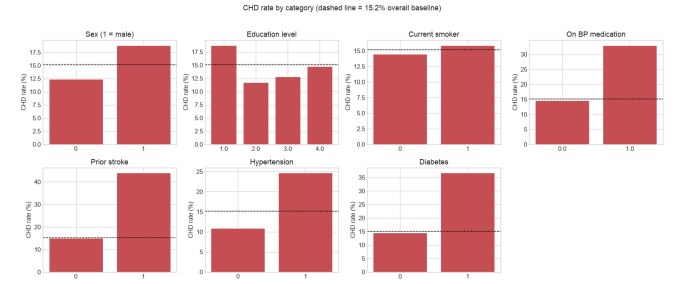


Figure IV: CHD rate within each categorical feature

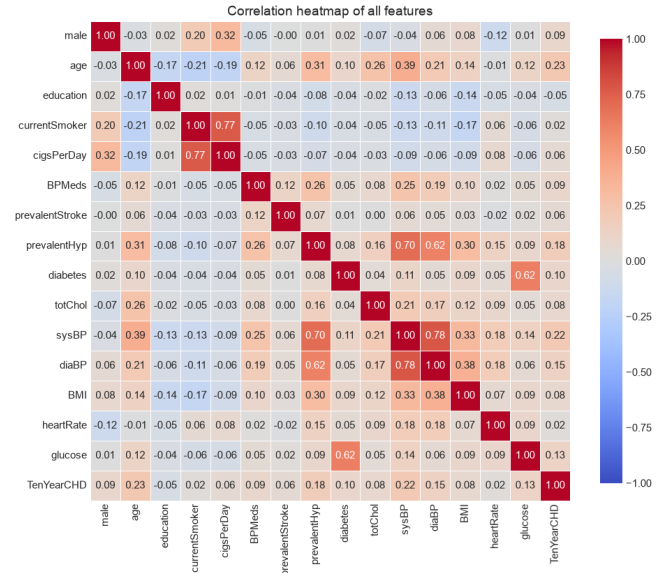


Figure V Correlation heatmap of all features

IV. METHODOLOGY

A. Preprocessing

When checking the missing data, we saw total empty rows of about 13.7% of the data so we didn't drop the data and proceeded to impute the data. On the preprocessing phase median imputation was done for the continuous data over the mean due to the right-skewness of the data and then did the StandardScaler, making the mean 0 and standard deviation of 1. For the binary kind of data we did the mode imputation, the one class that's most frequent and did no scaling, cause all the data are either 0 or 1. For categorical data, which is just one column "education" we did mode imputation same as for binary and then one-hot encoding for each, category converted it to the column and give it the binary values, 1 on which category the person belonged, making the total number of columns to 18. All this preprocessing of the data was done on the leakage safe design. This ColumnTransformer inside a pipeline was fitted on a training data only. All the data were categorized on four different classes; each category data was separated and was preprocessed according to the category. Target column for the train data was dropped and split was made on the ratio of 80/20.

B. Class Imbalance Handling

The other problem on the dataset was the class imbalance which was seen while doing the EDA. Keeping in

mind that we were going to run the model on 5 different classical and 2 ensembles methods we had two choices, one was to use the class weights, giving the weights for the minority class more weights and the other was the Synthetic Minority Over-Sampling Technique (SMOTE) which rebalances the data by generating synthetically the minority class data through interpolation between a minority sample and its nearest minority class neighbors [10]. Algorithm such as Naive Bayes and k-NN didn't had the class weights parameter so it was a natural choice to move on with the SMOTE method for rebalancing the classes, in this way we are able to avoid giving some model an imbalance-handling advantages. And a important thing to note here is that, we did the SMOTE operation inside the pipeline only, effectively showing it only to the training data, so that when the data is generated, we don't leak the test data as shown in the Figure VI its sits between the ColumnTransform and the classifier.

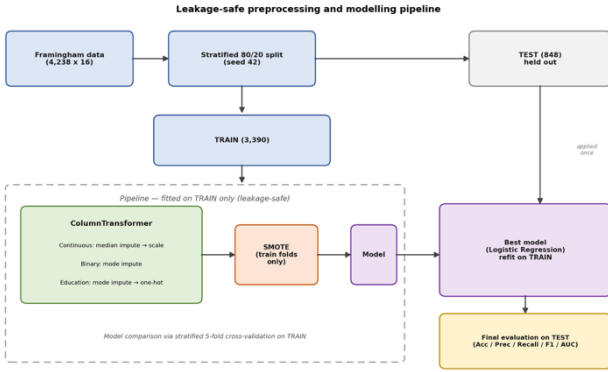


Figure VI Leakage safe preprocessing and modeling pipeline

C. Machine-Learning Algorithms

After the preprocessing, model was done on five classical machine learning algorithms and two ensemble algorithms. The first one was the Logistic Regression – a linear type model, that predicts the probability via a linear combination of the features. Second one was the Naïve Bayes (Gaussian) it's a probabilistic algorithm that applies Bayes' theorem assuming feature are independent. Since we have a continuous features GaussianNB variant algorithm was used which assumes the normal distribution per feature. The third type of algorithm was the distance-based algorithm, k-NN we set the hyperparameter $k = 5$ enabling to look for the 5 nearest neighbors, as it classifies classes distance based: votes the majority class. The fourth type is a tree based: Decision tree two ensemble Random Forest and Gradient Boosting. Decision tree splits the data no feature thresholds to form rules, interpretable but prone to overfitting. Random forest combines many decision trees via bagging(averaging), reducing a single tree's instability. And Gradient Boosting which combines the multiple trees sequentially, each tree correcting the previous one's errors. And finally, the SVM – margin-based approach, we used the RBF kernel variant for our data. SVM finds a maximum-margin boundary, and the RBF kernel lets that boundary be non-linear. And for the hyperparameter we didn't change that much, max iteration was set to 2000 for logistic regression and random state was set to 42 for reproducibility and rest of the model was on default setting. All the above-mentioned algorithms were implemented in scikit-learn and imbalanced-learn for SMOTE.

D. Comparative Experiment: Feature Scaling

After all models were run with the hyperparameters described above and every evaluation metric measured with scaling **on**, the identical procedure was repeated with scaling **off** the same data, the same split, the same SMOTE, the same seven models, and the same 5-fold cross-validation, changing nothing except the removal of standardization from the continuous features. This controlled setup isolates feature scaling as the single variable and directly addresses the research question: how does standardization affect model performance? We hypothesized that the largest changes would occur for the distance- and gradient-based methods (k-NN and SVM), which rely on feature magnitudes; little to no change for the tree-based methods (Decision Tree, Random Forest, Gradient Boosting) and the probabilistic Naïve Bayes, which do not compare features by scale; and only a mild effect, if any, for Logistic Regression. To judge whether an observed change was meaningful rather than noise, each model's scaled-versus-unscaled difference in AUC was compared against its cross-validation fold-to-fold standard deviation; a change was treated as a real effect only when it exceeded that standard deviation.

E. Evaluation Metrics

Evaluating the model is essential to understand how well it performs and where it may need tuning. As this is a classification task, the following evaluation metrics were used: Accuracy, Precision, Recall, F1-score, Confusion Matrix, and AUC. Accuracy gives the overall fraction of correct predictions, but it can be misleading under class imbalance, as it overstates models that largely ignore the minority class; for this reason, recall, F1-score, and AUC were treated as the primary metrics. Precision measures, of the cases the model predicts as positive, how many are truly positive (the false-alarm cost). Recall (sensitivity) measures, of the true CHD cases, how many the model correctly identifies; it is prioritised here because a missed at-risk patient (a false negative) is clinically more costly than a false alarm. F1-score is the harmonic mean of precision and recall, balancing the two. The confusion matrix gives the raw breakdown of true positives, false positives, false negatives, and true negatives, from which the other metrics are derived. Finally, AUC (ROC) is a threshold-independent measure of ranking ability that is robust to class imbalance; because it does not depend on the 0.5 decision threshold, it was also used as the metric for assessing the feature-scaling effect. AUC is equivalent to the clinical C-statistic, the same measure reported for the Framingham risk score (approximately 0.76), providing a direct point of comparison with established clinical practice.

V. EXPERIMENTAL SETUP

This section describes the experimental setup. The complete data was divided using 80/20 train-test split (3390 / 848) and set the random seed 42 for the reproducibility. 5-fold cross-validation on the training set for model comparison and selection; the held-out test was used once, to evaluate final selected model. For the decision threshold it was set to 0.5. The overall experiment was implemented in Python and its libraries with scikit-learn and imbalanced-learn classifiers used default hyperparameter unless stated.

VI. RESULTS

A. Model Comparison

Here we will compare and present all the models evaluation metrics on 5 -fold CV.

TABLE III Model Comparison (5-fold cross validation, scaling ON, ranked by AUC)

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.673	0.269	0.670	0.384	0.725
Gradient Boosting	0.790	0.324	0.351	0.337	0.700
Naive Bayes	0.776	0.312	0.392	0.347	0.694
Random Forest	0.819	0.338	0.200	0.250	0.683
SVM	0.693	0.245	0.485	0.325	0.659
k-NN	0.655	0.234	0.559	0.330	0.632
Decision Tree	0.731	0.221	0.307	0.257	0.557

The clear winner based on AUC and recall and F1 here is Logistic Regression with scaling enabled. LR performed best overall on the experiment as shown on the Figure VII and VII

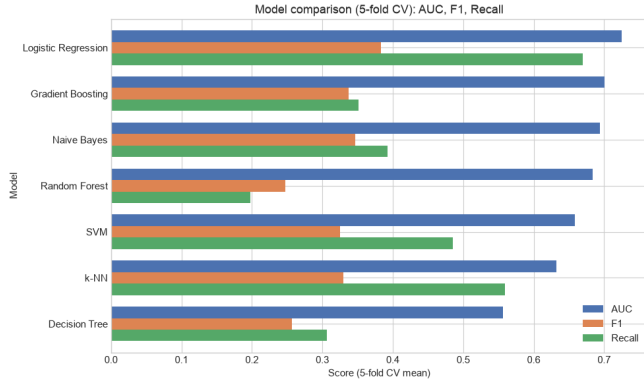


Figure VII Model Comparison (AUC, F1, Recall)

B. Feature-Scaling Experiment

TABLE IV Effects of feature scaling on AUC(5- fold CV).

Model	AUC (ON)	AUC (OFF)	Change	Fold std	Effect
Logistic Regression	0.725	0.727	-0.002	0.028	none (< noise)
Naive Bayes	0.694	0.707	-0.013	0.032	none (< noise)
k-NN	0.632	0.608	+0.025	0.022	marginal
Decision Tree	0.557	0.555	+0.002	0.021	none (< noise)
SVM	0.659	0.726	-0.067	0.026	REAL (~2.5× noise)
Random Forest	0.683	0.687	-0.004	0.011	none (< noise)
Gradient Boosting	0.700	0.706	-0.005	0.026	none (< noise)

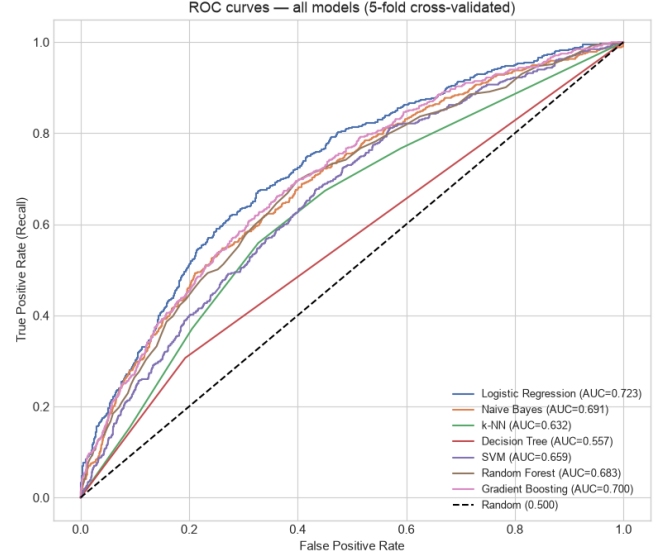


Figure VIII ROC curve for all model (CV)

As shown in Table VI and Fig. IX, feature scaling significantly affected only the SVM. For every other model, the change in AUC between the scaled and unscaled conditions was smaller than the cross-validation fold standard deviation and was therefore treated as noise. The SVM was the sole exception: its AUC changed by roughly 2.5 times the fold standard deviation, and — contrary to the common expectation that distance-based methods benefit from scaling — the *unscaled* SVM performed better. k-NN showed only a marginal change, and all remaining models were unaffected.

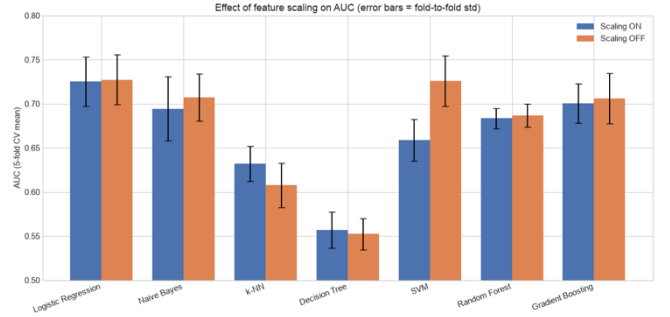


Figure IX Effect of feature scaling on AUC per model

C. Final Model Evaluation (Test Set)

Final model (Logistic Regression, held-out test set, threshold 0.5): Accuracy 0.654; Precision 0.244; Recall 0.605; F1 0.347; AUC 0.693. Confusion matrix: TN 477, FP 242, FN 51, TP 78 : 78 of 129 at-risk patients correctly identified on CV while the test-set score (AUC 0.693, recall 0.605) closely matched the CV estimated values which is reassuring as it indicated the model have generalized and is not over fitted as shown on the Figure X.

Final model(LR) confusion matrix (test set, threshold 0.5)

Actual	No CHD	477	242
	CHD	51	78
	Predicted	No CHD	CHD

Figure X Confusion Matrix of the final model (Logistic Regression) on the test set.

VII. DISCUSSION

The strongest supported conclusion is that Logistic Regression performs best on AUC, recall, and F1 in the scaled comparison, and is nearly tied with SVM on the unscaled AUC. Adding ensemble complexity did not improve the primary metrics under these configurations. This does not, however, establish that the outcome relationship is predominantly linear: weak pairwise correlations and a stronger linear model are insufficient to identify the underlying functional form. Selecting the scaled Logistic Regression pipeline is reasonable its scaling difference is negligible (-0.002) and it leads the comparison but we did not prespecify a formal rule for choosing the final scaling condition, and an explicit validation-based selection rule would reduce the risk of retrospective justification.

The scaling experiment intervenes on the whole preprocessing SMOTE classifier pipeline. Because SMOTE relies on neighbour distances, and those distances change under standardization, the synthetic training samples can differ between conditions even at a fixed seed; small changes for tree-based models are therefore plausible despite their usual scale-insensitivity. For the SVM, the default $\gamma = 1/\text{scale}$ depends on input variance, so standardization alters the effective kernel width as well as the oversampling neighbourhoods. The observed unscaled advantage should therefore not be generalized into a recommendation against scaling SVMs; a focused follow-up would tune C and gamma within nested validation and examine resampling separately.

Several limitations bound these findings. Results come from a single 80/20 split with fixed seeds, so the rankings are not shown to be stable across other splits or software versions; confidence intervals or repeated validation would help distinguish the small differences observed. The dataset is an anonymized teaching subset intended for educational use, so this work should be read as a methodological exercise rather than a clinical study external validity, subgroup performance, and prospective use were not assessed. Finally, because oversampling changes the training class distribution, the model's outputs should not be read as

calibrated ten-year probabilities; calibration and decision costs were not evaluated, and the evidence supports neither deployment nor a claim of clinical benefit. Future work should compare imbalance treatments (including none), tune key parameters and select an operating threshold within the training data, and assess calibration recording each extension as a new experiment rather than a result already established here.

VIII. CONCLUSION

For the Framingham teaching-data experiment, Logistic Regression offered the strongest balance of ROC-AUC, recall, and F1 among the seven scaled pipelines, and notably outperformed both ensemble methods, indicating that added model complexity provided no benefit on this weak, largely linear signal. Its mean validation AUC was 0.725 and its held-out test AUC 0.693; at the default threshold it identified 78 of 129 positive cases (recall 0.605) while producing 242 false positives supporting its use as a baseline for further investigation rather than as a validated screening system.

The feature-scaling comparison showed that preprocessing can alter model rankings: SVM exhibited the largest difference, improving from 0.659 with scaling to 0.726 without it, while all other models were unaffected beyond cross-validation noise. Because standardization also affects SMOTE's neighbor distances and the default SVM kernel parameter, the experiment does not isolate a universal effect of scaling on the classifier alone. In answer to the research question, then, Logistic Regression was the best-performing algorithm and feature scaling materially affected only the SVM. More controlled validation, documented dataset provenance, and complete reproducibility records are needed before stronger conclusions can be drawn.

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